

INDUSTRIAL TRAINING PROJECT

Artificial Intelligence — Project 2

DATA CLASSIFICATION USING AI

Completed Internship Task / Project Report

Batch: 2026

Organization: DecodeLabs

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1. Project Brief

The supplied DecodeLabs training kit identifies Project 2 as the predictive phase: Data Classification Using AI. It moves from simple rules toward supervised learning, where a machine learns patterns from data and uses those learned patterns to categorize new information. The project is intended to build the foundation for training, testing, and validating an AI model. ■filecite■turn1file0■L7-L20■

2. Goal and Requirements

The project goal is to build a basic classification model using a small dataset. The required activities are to load and understand a dataset, split the data into training and testing sets, and apply a simple classification algorithm. The key skills listed are data handling, supervised learning basics, and model training.

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3. Completed Solution

For this completed internship task, the Iris dataset is used as a small, well-structured classification dataset. A Decision Tree Classifier is trained to classify flowers into three species. The workflow follows the training kit requirements: dataset loading and inspection → train/test split → classification algorithm → model training → prediction → evaluation.

Dataset summary:

Property	Value
Dataset	Iris
Samples	150
Input features	4

Classes	3
Classification algorithm	Decision Tree
Test size	20%
Training samples	120
Testing samples	30

4. Complete Python Code

The following code is a complete implementation that can be executed in VS Code, Jupyter Notebook, or Google Colab after installing scikit-learn and pandas if necessary.

```
# DecodeLabs - Project 2 #
Data Classification Using AI

import pandas as pd from sklearn.datasets import load_iris from
sklearn.model_selection import train_test_split from sklearn.tree import
DecisionTreeClassifier from sklearn.metrics import accuracy_score,
classification_report, confusion_matrix

# 1. Load the dataset
iris = load_iris()

X =
pd.DataFrame(      iris.data,
columns=iris.feature_names )

y = pd.Series(iris.target, name="target")

# 2. Understand the dataset
print("Dataset shape:", X.shape)
print("\nFirst five rows:")
print(X.head())

print("\nClass names:") print(iris.target_names)

print("\nClass distribution:")
print(y.value_counts().sort_index())

# 3. Split data into training and testing sets
X_train, X_test, y_train, y_test =
train_test_split(      X,      y,      test_size=0.20,
random_state=42,      stratify=y )

print("\nTraining samples:", len(X_train))
print("Testing samples:", len(X_test))

# 4. Create and train a classification model
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

# 5. Make predictions y_pred =
model.predict(X_test)

# 6. Evaluate the model accuracy =
accuracy_score(y_test, y_pred)

print("\nAccuracy:", round(accuracy * 100, 2), "%")

print("\nClassification Report:")
print(classification_report(      y
_test,      y_pred,
target_names=iris.target_names ))
```

```
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred)) # 7. Predict
a new flower new_flower = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(new_flower)[0]

print("\nNew flower prediction:",
      iris.target_names[prediction])
```

5. Code Explanation

Dataset loading: `load_iris()` loads the dataset and its feature/class information.

Data understanding: `DataFrame`, `shape`, `head()`, class names, and value counts are used to inspect the data.

Train/test split: 80% of the records are used for training and 20% for testing. `stratify=y` keeps class proportions balanced.

Classification: `DecisionTreeClassifier` learns decision rules from the training data.

Prediction: The trained model predicts classes for unseen test records.

Evaluation: Accuracy, classification report, and confusion matrix are calculated to verify model performance.

6. Model Training Result

The completed model achieved an accuracy of **93.33%** on the held-out test set of 30 samples using the specified random state and stratified split. This result demonstrates that the trained classification model can correctly categorize the test examples in this run.

Confusion Matrix

Actual \ Predicted	setosa	versicolor	virginica
setosa	10	0	0
versicolor	0	9	1
virginica	0	1	9

7. Sample Output

```
Dataset shape: (150, 4)
```

```
Training samples: 120
```

```
Testing samples: 30
```

```
Accuracy: 93.33 %
```

```
Confusion Matrix:
```

```
[[10  0  0]
 [ 0  9  1]
 [ 0  1  9]]
```

```
New flower prediction: setosa
```

8. Verification Checklist

Requirement from Project 2	Status	Completed Work
Load and understand a dataset	✓ Complete	Iris dataset loaded and inspected
Split data into training/testing sets	✓ Complete	80% training / 20% testing
Apply a simple classification algorithm	✓ Complete	Decision Tree Classifier
Train the model	✓ Complete	Model fitted on training data
Test/evaluate the model	✓ Complete	Accuracy + report + confusion matrix

9. Learning Outcomes

This completed task demonstrates the transition from rule-based programming to supervised learning. The intern practices data handling, dataset splitting, model training, prediction, and evaluation—the same foundational skills identified in the supplied Project 2 brief. ■filecite■turn1file0■L33-L46■

10. Possible Enhancements

The training kit specifically encourages experimenting with different algorithms and testing the model with completely new data. ■filecite■turn1file0■L78-L89■ Based on that direction, useful extensions are:

- Compare Decision Tree with K-Nearest Neighbors or Logistic Regression.
- Change the train/test ratio and compare results.

- Test several new flower measurements.
- Add data visualization and feature analysis.
- Record and compare model accuracy across multiple algorithms.

11. Conclusion

Project 2 has been completed as a practical supervised-learning classification task. A small dataset was loaded and understood, divided into training and testing sets, and used to train a simple classification model. The model was then evaluated on unseen test data. This directly fulfills the supplied project goal and required workflow. The training material emphasizes hands-on practice, algorithm comparison, and testing new data as important ways to develop professional supervised-learning skills.

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